

F25_T06 – Technology Requirements & Learning Plan

DriveTree: Book · Learn · Go

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A) Technological Requirements

This section provides a detailed breakdown of all technologies planned for the DriveTree application, including purpose, justification, advantages, limitations, and alignment with Sprint 1–3 documentation.

A.1 Planned Technologies Overview

Category	Technology	Purpose	Pros	Cons
Frontend	React (Vite)	Build a fast SPA for Students, Instructors, Admin; supports routing & reusable components.	Industry standard; fast dev server; reusable components; strong ecosystem; works with Tailwind.	Requires state management knowledge; routing setup complexity.
Styling	Tailwind CSS	Provides utility-first styling for consistent color, spacing, layout.	Speeds up UI creation; maintains design consistency; mobile-friendly.	HTML becomes class-heavy; requires understanding of utility classes.
Backend Runtime	Node.js	Run backend JavaScript code using non-blocking, event-driven I/O.	Fast, scalable; same language as frontend; huge library support.	Single-threaded limitations; debugging async code can be challenging.
Backend Framework	Express.js	REST API framework for authentication, bookings, reviews, instructor verification.	Simple, minimalistic; flexible routing; large community; lightweight.	No strict structure enforced; requires manual setup of middleware.
Database	MongoDB Atlas	Cloud-hosted NoSQL DB for users, bookings,	Flexible schema; auto-scaling; free tier; easy to	No strict relationships; must manually enforce

		instructors, reviews.	integrate with Node.	data design rules.
Authentication	JWT	Secure token-based login for Student/Instructor/Admin roles.	Stateless; ideal for SPA; widely used; simple to implement.	If token leaked, risk of misuse; requires proper expiration strategy.
Cloud Storage	Firebase Storage / Cloudinary	Store instructor license files, profile images, lesson attachments.	CDN optimized; fast uploads; reduces DB load.	Separate integration; permission rules must be configured carefully.
Deployment	Render / Railway (Free Tier)	Deploy backend and frontend to cloud environment using GitHub integration.	Simple deployment; environment variables; logs included.	Free tier has sleep mode; limited monthly runtime.
Version Control	Git + GitHub	Collaborative repository for managing codebase and documentation.	Branching, PR reviews, change tracking, version history.	Merge conflicts if workflow not followed strictly.
Prototype	Figma	Full UI/UX mockup with interactive flows for Sprint 4 presentation.	Free education plan; real-time collaboration; easy prototyping.	Some advanced features are locked in free tier.
API Testing	Postman	Test endpoints for login, CRUD operations, booking flows.	Easy to use; supports collections; useful during backend build.	Team members must learn request structuring.
Optional	Docker	Create consistent dev environments for backend and MongoDB.	Portable; avoids environment mismatch issues.	Adds extra complexity for newcomers; optional for Sprint 4.

B) Learning Plan

The following tables describe each team member's technical responsibilities, current skill level, and planned learning trajectory aligned with Sprint 4 requirements.

Heemal Syangbo

Skill	Responsibility	Current Skill	Learning Timeline	Notes
React (Advanced + Optimization)	Main UI development	75%	Oct 2025 – Feb 2026	Semester 3 & 5 already covered React basics. Now improving for Capstone II.
Tailwind CSS	Full UI styling	65%	Oct 2025 – Jan 2026	Will master reusable components for DriveTree.
React Router + State Mgmt	Navigation + booking flow	60%	Nov 2025 – Feb 2026	Needed for multi-role (Student/Instructor/Admin) navigation.
API Integration (Axios)	Connect FE with BE	70%	Dec 2025 – Feb 2026	Integration starts after backend stabilizes.
Deployment (Vercel/Render)	Frontend hosting	50%	Feb 2026 – Mar 2026	Deployment done during Capstone II.

Jeffin Yohannan

Skill	Responsibility	Current Skill	Learning Timeline	Notes
Figma (Advanced Prototyping)	Full UI/UX for Sprint 4	80%	Sept 2025 – Nov 2025	Finishes Sprint 4 mockup.
Interaction Design	User flow mapping	70%	Sept 2025 – Dec 2025	Already skilled due to Sem 3/4 courses.
Accessibility UI	Clean & simple UI	50%	Jan 2026 – Mar 2026	Required for public-facing app.
React Basics	Assist FE dev	45%	Jan 2026 – Apr 2026	Completes practical React tasks in Capstone II.

Anudhin Thomas

Skill	Responsibility	Current Skill	Learning Timeline	Notes
Node.js (Advanced)	Backend logic	70%	Oct 2025 – Feb 2026	Already learned web backend in Sem 3/5. Now polishing for production readiness.
Express.js	Full API layer	65%	Nov 2025 – Feb 2026	Build Instructor verification, bookings, reviews.
MongoDB Atlas	Database design	60%	Nov 2025 – Mar 2026	Will design schemas, indexes & relationships in Capstone II.
JWT + Role-Based Auth	Authentication	55%	Dec 2025 – Feb 2026	Needed for Student/Instructor/Admin/Verified roles.
Deployment (Render/Railway)	Backend hosting	45%	Feb 2026 – Mar 2026	Final deployment preparation for April showcase.

Nashiruddin Feroz

Skill	Responsibility	Current Skill	Learning Timeline	Notes
Figma (Advanced Prototyping)	Full UI/UX for Sprint 4	95%	Sept 2025 – OCT 2025	Finishes Sprint 4 mockup.
Manual Testing (UI + API)	Full QA ownership	75%	Oct 2025 – Apr 2026	QA experience from COMP2136 (Sem 3).
Postman API Testing	Backend testing	60%	Nov 2025 – Feb 2026	Will test every API (booking, login, reviews).
MongoDB Query Basics	DB verification	50%	Jan 2026 – Mar 2026	Needed for checking booking & user data.
Documentation	Reports, test logs	85%	Ongoing	Handles all reports for Sprint 4, 5, 6.